

Trainings ▾

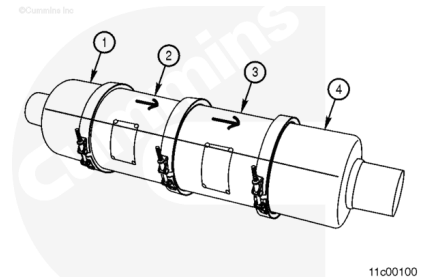
General Information

⚠ WARNING ⚠

During regeneration, exhaust gas temperature could reach 800°C [1500°F], and exhaust system surface temperature could exceed 700°C [1300°F], which is hot enough to ignite or melt common materials, and to burn people. The exhaust and exhaust components can remain hot after the vehicle has stopped moving. To avoid the risk of fire, property damage, burns or other serious personal injury, allow the exhaust system to cool before beginning this procedure or repair and make sure that no combustible materials are located where they are likely to come in contact with hot exhaust or exhaust components.

⚠ CAUTION ⚠

The catalyst elements contained in the aftertreatment system are made of brittle material. Do not drop or strike the side of the aftertreatment system as damage to the catalyst element can result.



Due to the number of various exhaust aftertreatment applications, this procedure has been written to be generic. Illustrations within this procedure will **not** represent all applications.

The aftertreatment system is composed of four sections. These sections are:

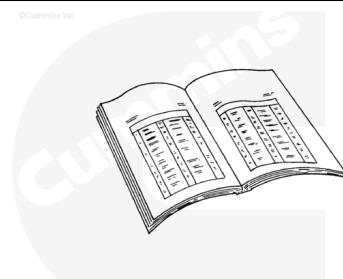
1. Inlet
2. Aftertreatment diesel oxidation catalyst
3. Aftertreatment diesel particulate filter
4. Outlet.

Note : In some applications, the catalyst can be integrated into the inlet of the exhaust aftertreatment system.

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



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⚠ CAUTION ⚠

The aftertreatment diesel particulate filter differential pressure sensor will not operate properly if the differential pressure sensor tubes are not connected to the correct port. Mark the differential pressure sensor tube connection locations before disconnecting.

- Disconnect the batteries. Refer to Procedure 013-009 (Battery Cables and Connections) in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Mark the direction of exhaust flow to aid in assembly.
- Draw an orientation reference line across each of the v-band clamps, aftertreatment canister sections, and connection points to the tailpipe. This will aid in returning the sections and the v-band clamps to their original orientation during installation.
- Mark the number of each exhaust gas temperature sensor connector prior to disconnecting the exhaust temperature sensor from the wiring harness.
- Disconnect the aftertreatment gas temperature sensor wiring harness connectors. Refer to Procedure 019-449 (Aftertreatment Gas Temperature Sensor) in Section 19 of the Troubleshooting and Repair Manual, CM871 and CM876 Electronic Control Systems, ISX and ISM Engines, Bulletin 4021560. (</qs3/pubsys2/xml/en/procedures/16/16-019-449.html>)
- Remove the mounting straps or bolts from the inlet, if necessary. Refer to the OEM troubleshooting and repair manual.

- Remove the mounting straps or bolts from the outlet, if necessary. Refer to the OEM troubleshooting and repair manual.
- Disconnect the aftertreatment diesel particulate filter differential pressure sensor tube from the outlet, if necessary. Refer to Procedure 011-047 (Aftertreatment Diesel Particulate Filter Differential Pressure Sensor Tubes) in Section 11.
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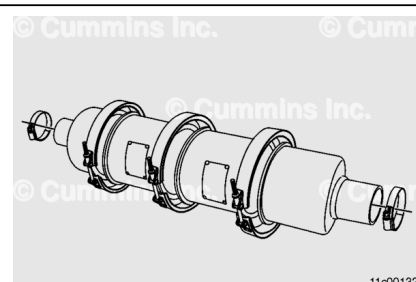
Remove

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ CAUTION ⚠

The catalyst elements contained in the aftertreatment system are made of brittle material. Do not drop or strike the side of the aftertreatment system as damage to the catalyst element can result.



In some applications, the catalyst is integrated into the aftertreatment inlet. Refer to Procedure 011-046 (Aftertreatment Diesel Particulate Filter Differential Pressure Sensor Mounting Bracket) in Section 11 for removal of the differential pressure sensor mounting bracket.
(/qs3/pubsys2/xml/en/procedures/101/101-011-046-tr.html)

Support the catalyst and aftertreatment diesel particulate filter to avoid applying a cantilevered load on the v-band joints.

Remove the Torca™ or v-band clamp used to connect the aftertreatment system to the exhaust pipe, if necessary.

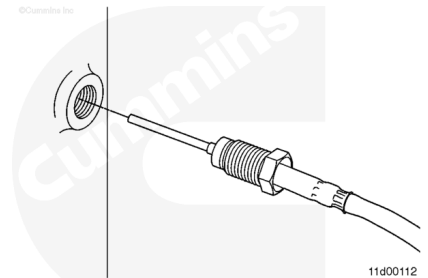
Remove the v-band clamp connecting the inlet to the catalyst, if necessary.

Remove the v-band clamp connecting the outlet to the aftertreatment diesel particulate filter, if necessary.

Remove and discard the gaskets.

Disassemble

Remove the aftertreatment gas temperature sensor from the boss in the inlet and/or outlet, if necessary. Refer to Procedure 019-449 (Aftertreatment Gas Temperature Sensor) in Section 19 of the Troubleshooting and Repair Manual, CM871 and CM876 Electronic Control Systems, ISX and ISM Engines, Bulletin 4021560. (</qs3/pubsys2/xml/en/procedures/16/16-019-449.html>)



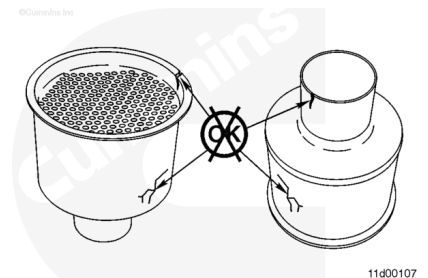
Clean and Inspect for Reuse

Inspect the aftertreatment inlet and outlet canisters for cracks or other damage.

Inspect the aftertreatment gas temperature sensor boss threads for damage, if the sensor was removed.

Inspect the aftertreatment diesel particulate filter differential pressure sensor tube boss threads for damage, if the tube was removed.

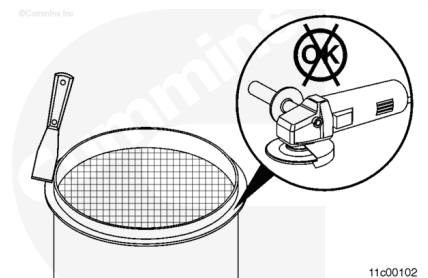
If thread damage is found on the aftertreatment gas temperature sensor boss threads or differential pressure sensor tube boss threads, a helicoil **must** be used for repair.



Flange Preparation - Use a putty knife to remove any residual gasket material from the flanges on the aftertreatment diesel particulate filter.

Avoid dropping fragments of gasket material into the aftertreatment diesel particulate filter.

Do **not** use compressed air to remove fragments of gasket material that have fallen into the filter during gasket removal.



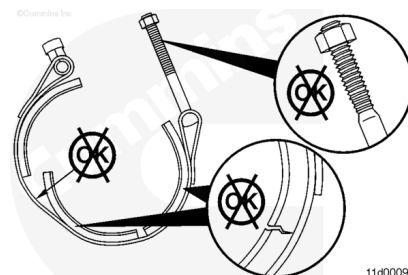
Do **not** grind on the flange surface, as this can damage the flange and cause the connection to leak.

Inspect the v-band clamps and mounting straps for signs of over-extension.

The band **must not** be bent or damaged.

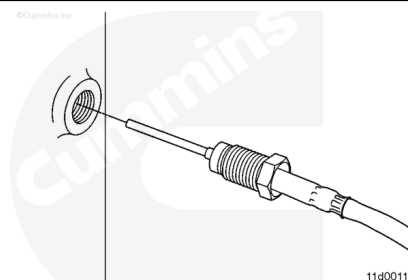
Inspect the v-band clamp and mounting strap threads for damage.

Replace the v-band clamp or mounting strap if damage is found.



Assemble

Install the aftertreatment gas temperature sensor in the boss on the inlet and/or outlet. Refer to Procedure 019-449 (Aftertreatment Gas Temperature Sensor) in Section 19 of the Troubleshooting and Repair Manual, CM871 and CM876 Electronic Control Systems, ISX and ISM Engines, Bulletin 4021560. (</qs3/pubsys2/xml/en/procedures/16/16-019-449.html>)

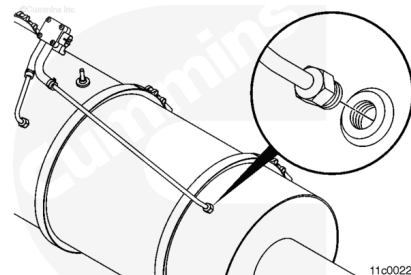


Install

Apply a coat of anti-seize compound on the treads of the v-band clamps and Torca™ clamps.

The aftertreatment outlet section contains a differential pressure sensor tube boss. Align the differential pressure sensor tube boss with the differential pressure sensor tube nut prior to tightening the v-band clamp or the Torca™ clamp.

Note : If the Torca™ clamp is replaced during service, be sure to replace it with another Torca™ clamp. Do not use a u-bolt as a replacement clamp. U-bolt clamps can crush the tailpipe and make it difficult to remove the aftertreatment system for future service.



Tighten the Torca™ clamps or v-band clamps used to secure the inlet and outlet of the aftertreatment to the exhaust pipe.

Torque Value:

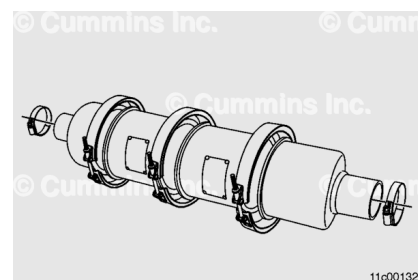
Torca™ Clamp

1. 55 n•m [41 ft-lb]

Torque Value:

V-Band Clamp

1. 14 n•m [124 in-lb]



Install a new gasket between the v-band clamp joints that have been disconnected.

Install the v-band clamp used to connect the outlet to the aftertreatment diesel particulate filter, if necessary.

Install the v-band clamp use to connect the inlet to the catalyst, if necessary.

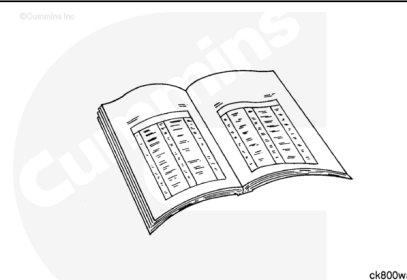
Tighten the v-band clamps.

Torque Value: 20 n•m [177 in-lb]

Finishing Steps

⚠ WARNING ⚠

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⚠ CAUTION ⚠

If the temperature sensor wire connectors are not connected to the proper locations after installation, aftertreatment system damage can result.

⚠ CAUTION ⚠

The aftertreatment diesel particulate filter differential pressure sensor will not operate properly if the differential pressure sensor tubes are not connected to the correct port. Install the differential pressure sensor tubes as noted during disassembly.

- Install the mounting straps or bolts to the inlet, if necessary. Refer to the OEM troubleshooting and repair manual.
- Install the mounting straps or bolts from the outlet, if necessary. Refer to the OEM troubleshooting and repair manual.
- Connect the aftertreatment gas temperature sensor wiring harness connectors, if necessary. Refer to Procedure 019-449 (Aftertreatment Gas Temperature Sensor) in Section 19 of the Troubleshooting and Repair Manual, CM871 and CM876 Electronic Control Systems, ISX and ISM Engines, Bulletin 4021560.
(/qs3/pubsys2/xml/en/procedures/16/16-019-449.html)
- Connect the aftertreatment diesel particulate filter differential pressure sensor tube to the outlet, if necessary. Refer to Procedure 011-047 (Aftertreatment Diesel Particulate Filter Differential Pressure Sensor Tubes) in Section 11.
(/qs3/pubsys2/xml/en/procedures/101/101-011-047-tr.html)
- Connect the vehicle batteries. Refer to Procedure 013-009 (Battery Cables and Connections) in Section 13.
(/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)
- Operate the vehicle on a dynamometer or perform a road test with the engine at rated load for a minimum of 5 minutes to make sure the aftertreatment system is performing properly. Refer to Procedure For ISX engines, 010-024 (Air Leaks, Air Intake and Exhaust Systems) in Section 10 in the Signature™, ISX, and QSX15 Service Manual, Bulletin 3666239.
(/qs3/pubsys2/xml/en/procedures/10/10-010-024-tr.html) Refer to Procedure For ISM engines, 010-024 (Air Leaks, Air Intake and Exhaust Systems) in Section 10 in the ISM, ISMe, and QSM11 Engines Service Manual, Bulletin 3666322.
(/qs3/pubsys2/xml/en/procedures/35/35-010-024-tr.html)

Last Modified: 29-Apr-2009
